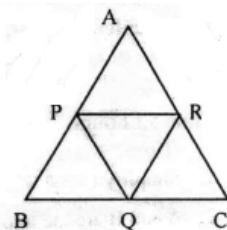


JMO – 2015

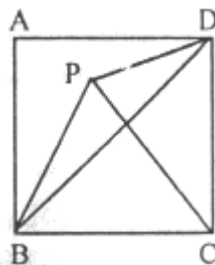
Answer all questions. All questions carry equal marks.

Full Marks 100

1. Simplify: $\sqrt{998 \times 999 \times 1000 \times 1001 + 1}$
2. A and B are the two extremities of a straight road. Ramesh is at A and Ranjan is at B. Each of them is to move by their car to other extremities and come back without wasting any time while turning back. On one occasion, Ramesh starting moving 30 minutes before Ranjan and they met at C which is 20 KM away from A. During their return journey. On another occasion, Ramesh started 15 minutes earlier than Ranjan and they met at D which is 30 KM away from A. If they drive with equal speed determine the distance between A and B.
3. There is an outlet pipe attached to the bottom of a tank and the inlet pipe pours 5 litres of water per minute more than that taken away by outlet pipe. If the outlet pipe is opened 30 minutes after opening the inlet pipe, then the empty tank becomes full in 32 minutes after the outlet pipe is opened. If the tank is half full when the outlet pipe is opened and the inlet pipe is opened 20 minutes after opening the outlet pipe, the tank gets filled in 52 minutes after the inlet pipe is opened. What is the capacity of the tank?
4. While continuing to do a work, A's capacity of doing work is decreased by 20% after every 4 hours of continuous work. On the other hand, while B continues to do a work, his capacity of doing work decreases by 25% after every 3 hours of continuous work. But in the beginning of doing a work, A and B with equal capacity. If A can complete the work in 3 hours, in what time B can complete it?
5. In the diagram below, ABC and PQR each is an equilateral triangle. If $\angle BPQ = 30^\circ$ and each side of triangle ABC is of length 12 cm, determine the area of triangle PQR.



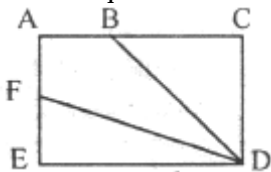
6. In the diagram below, ABCD is a square, and triangle PBC is an equilateral triangle. If BD is the diagonal of the square and each side of the square is $8\sqrt{3}$ cm long, determine the area of triangle PBD correct to one place of decimal (take $\sqrt{3} = 1.732$).



7. If a number is divided by 8, it leaves remainder 87, if it is divided by 7 leaves remainder 6, if it is divided by 6 leaves the remainder 5 and so on and lastly if it is divided by 2 leaves the remainder 1. Find such a number that lies between 7000 to 8000.
8. An exam hall with 50 rows of seats has 10 seats in the first row. Each successive row has one more seat than the previous one. If students taking an exam are permitted to sit in any row, but not next to another student in that exam hall, then find the number of students can be seated for an exam?

JMO QUESTIONS (ORISSA)

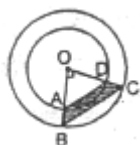
9. Namita usually leaves her cell phone on. If her cell phone is on but she is not actually using it, the battery will last for 24 hours. If she using it constantly, the battery will last for only 3 hours. Since the last recharge, her phone has been on for 9 hours, and during that time she has used it for 1 hour. If she does not talk any more but leaves the cell phone on, how many more hours will the battery last?
10. Rashmi brought three chocolates and four packets of hajmola for which he paid Rs. 2.65. Sheela brought one chocolate and two packets of hajmola for which she paid Rs. 1.05. Ani received 15 paise in change (the correct amount) when she gave Rs. 2/- to cover her purchase. What did Ani buy and how do you know what she brought?
11. A man on his way to a meeting shortly after 6.00 PM observes that the hands of his watch form an angle 110° . Returning before 7.00 PM, he notice that again the hands of his watch form an angle 110° . For how much time the man was out?
12. For a natural number ' n ' greater than one by the 'length' of the number we mean the number of factors in the representation of n as a product of prime numbers. For example the length of number 90 is 4, since $90 = 2 \times 3 \times 3 \times 5$. How many odd numbers between 2 and 100 have length 3?
13. How many 10 digit numbers of the form $abcdefghij$ (with $a = 1$ and all other digits equal to either 1 or 0), have the property that $a+c+e+g+i = b+d+f+h+j$?
- 14.
- (a) What is the smallest integer greater than 1 which is a square, a cube and also a fourth power?
- (b) What is the middle digit of the product $968880726456484032 \times 875$?
15. The rectangle shown has length $AC = 32$ units, width $AE = 20$ units, and B and F are midpoints of AC and AE respectively. Find the area of the quadrilateral ABDF.



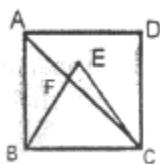
- 16.
- (a) Five straight lines intersect at a common points and five triangles are constructed as shown. What is total of the 10 angles marked on the diagram?



- (b) Area of ABCD is 25.9 cm^2 . Find the area of the ring.



17. ABCD is a square of area $1 + \sqrt{3}$ sq units. BEC is an equilateral triangle. AC meets BE at F, find the area of triangle BFC



JMO QUESTIONS (ORISSA)

18. In the given addition each digit replaced by a letter. If different letter represent different digit, then $C=?$

$$\begin{array}{r} A B C \\ A B \\ + A \\ \hline 300 \end{array}$$

19. Aman, Bharat, Chandan, Deepak, Eman, Fakir, Gopal and Hari were buying apples. One variety of apple cost Rs. 10/- for 11 and the other variety of apples cost Rs. 5/- for 7. After the purchases, four apples which had not paid for and rolled on to the floor, were picked up by someone not returned. As the eight people were leaving the shop, the shopkeeper stopped them and insisted on counting the apples they were carrying. He found that Aman had 33, Bharat 32, Chandan 78, Deepak 35, Eman 37, Fakir 36, Gopal 39 and Hari 34. Who seems to have the apples that had not been paid for?
20. Thirty eight children are seated equally spaced around a circle. They are numbered in order from 1 to 38. What is the number of the child opposite to child number 8?

